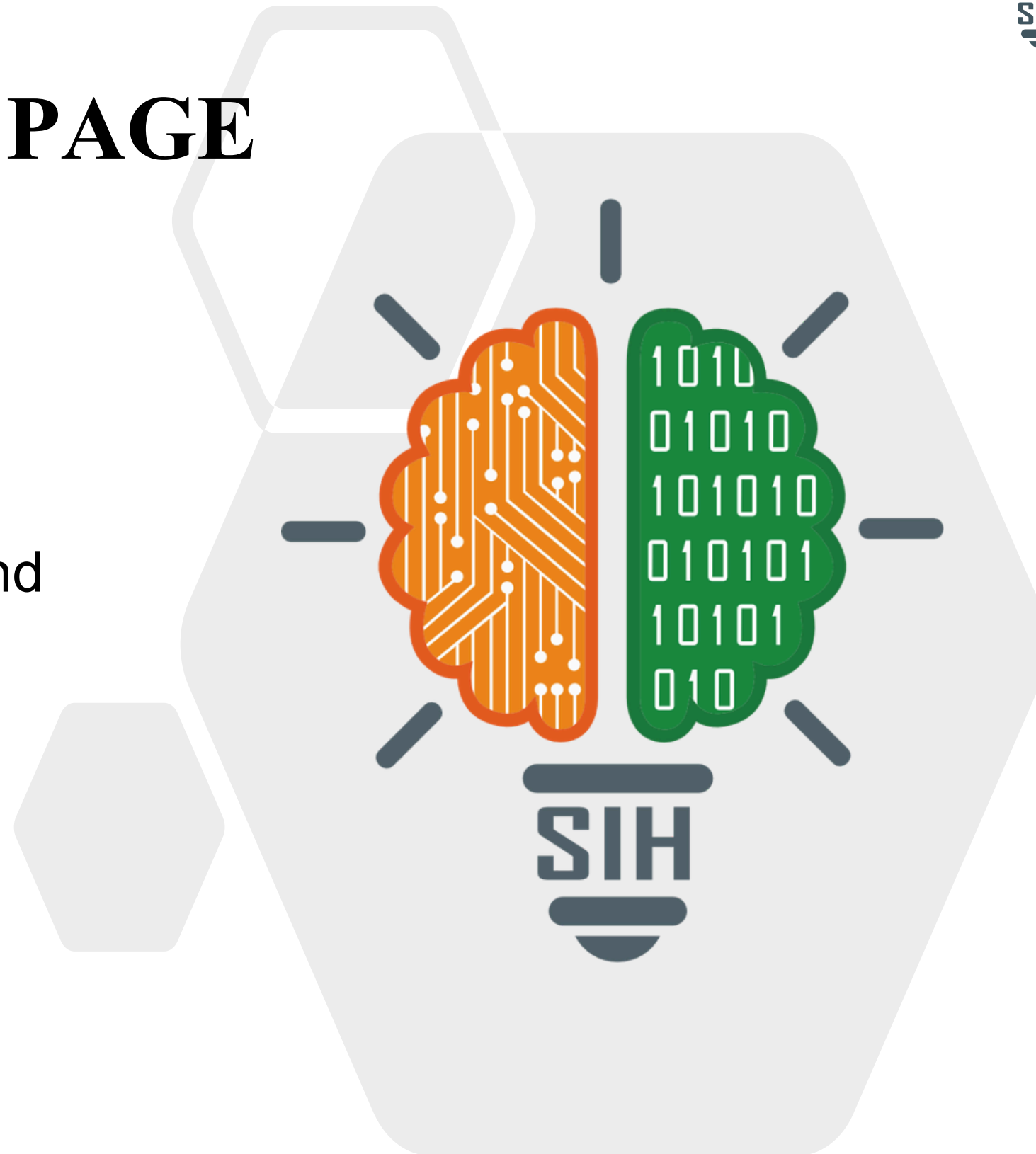




TITLE PAGE

- **Problem Statement ID** - SIH26107
- **Problem Statement Title** - AI-powered Intelligent Assistant for Indian Standards and BIS Services
- **Theme** - Smart Automation
- **PS Category** - Software
- **Team ID** -
- **Team Name** - localhost:8008



IDEA TITLE

Proposed Solution

Three independent AI-powered operating modes for Indian Standards & BIS services

KNOW

Standard Getter — AI Conversational Q&A

- **Multi-modal Input**
Text, voice & PDF input with multilingual support
— English + regional languages
- **Trustworthy FAQ Mode**
Answers grounded in official BIS/Indian Standards sources with citations
- **Smart Standards Search**
Find by IS Number, Part, Section, Year, Title, or Laboratory
- **AI Discovery Interview**
Guided conversational flow to identify applicable standards for any product

COMPLY

Producer Dashboard — Certification & Compliance Tracking

- **Compliance Checklist**
Step-by-step requirements auto-generated for each applicable standard
- **Certification Progress Tracker**
Real-time status dashboard for all active BIS applications
- **Smart Product Analyzer**
Upload brochure or PDF → AI identifies all applicable standards instantly
- **Form Pre-fill Assistant**
Conversationally collects info, auto-fills BIS application forms
- **What-If Compliance Simulator**
Test edge-case scenarios before final submission to avoid rejections

VERIFY

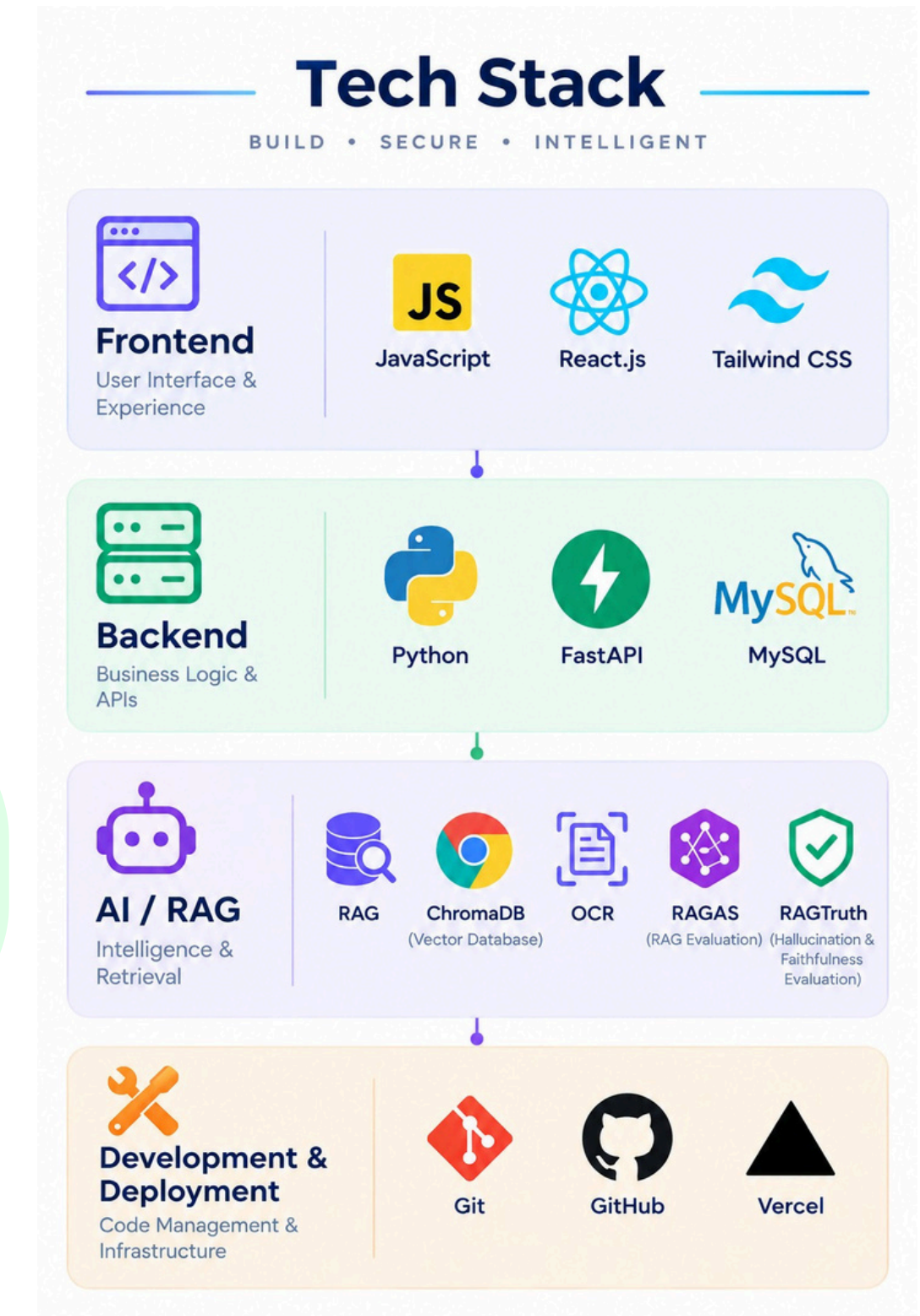
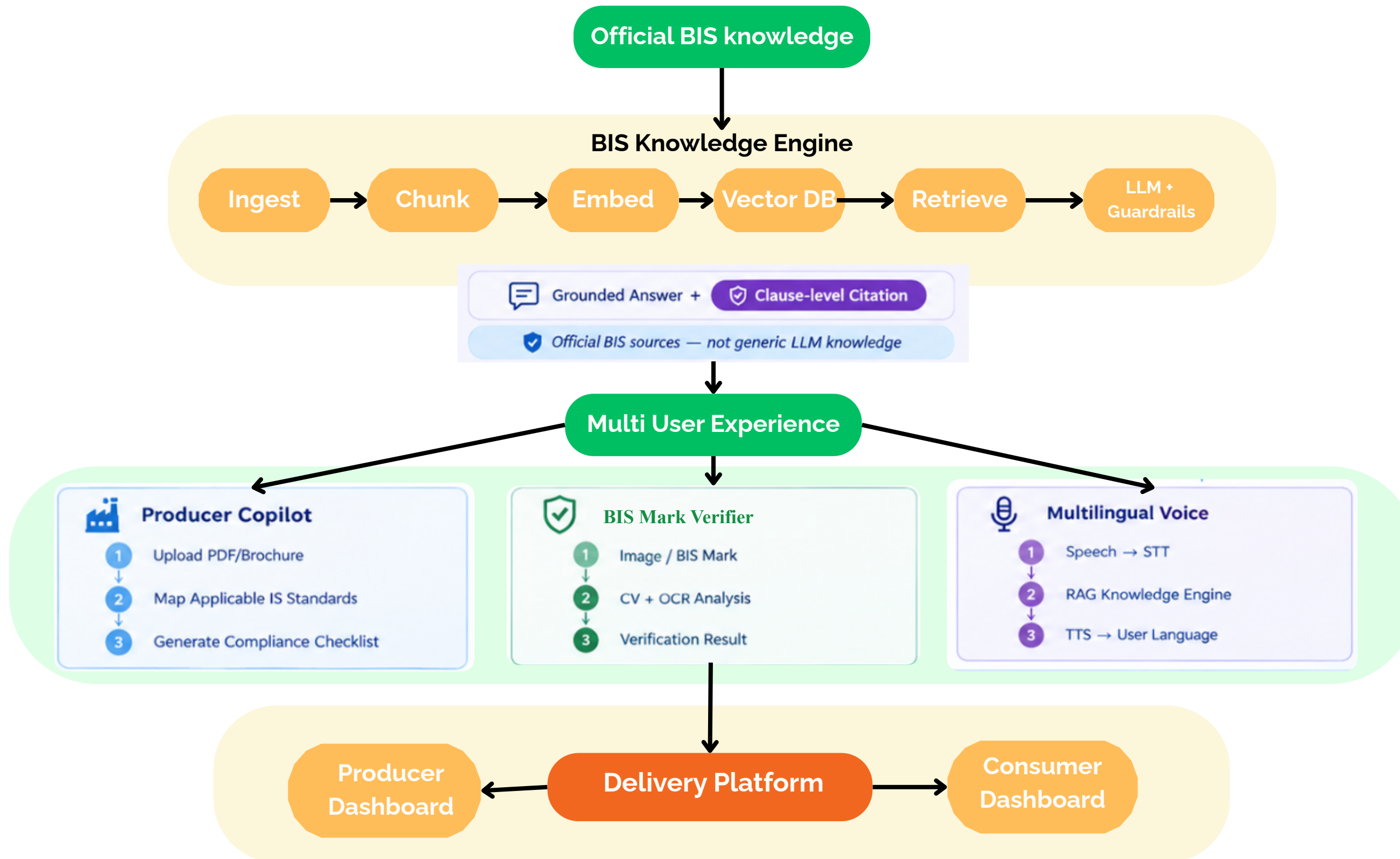
Consumer Dashboard — Trust & Authentication

- **BIS License Verification**
Instantly validate producer credentials against live BIS registry
- **HUID & Hallmarking Verification**
Check product authenticity by licence number or hallmark details
- **Hallmarking Status Lookup**
Real-time certification status query for hallmarked products
- **Fake BIS Mark Detector**
AI-powered detection and flagging of counterfeit ISI/BIS marks
- **AR Standard Overlay**
Camera scans product → identifies applicable Indian Standard + displays compliance info in real time

Result: Three separate user journeys — KNOW, COMPLY, and VERIFY operate independently as distinct modes.

localhost:8008

TECHNICAL APPROACH





FEASIBLE TODAY

- BIS portal, CARE app data, and IS standard documents are publicly available for knowledge base
- RAG + LLM is proven, production-ready technology, not experimental
- Can be launched as phased MVP rather than building everything at once



PHASED ROLLOUT

- Phase 1 (MVP): conversational Q&A, standards search, compliance checklist, certificate verification
- Phase 2 (Advanced): AR-based lookup, counterfeit detection, "what-if" compliance simulator
- Phased approach reduces delivery risk and proves core value before complex features



RISKS & MITIGATIONS

- Scattered PDFs and portals: automated crawler with change-detection keeps the knowledge base synced.
- Revised standards: version tracking and re-indexing keep answers aligned with the latest standard.
- Multilingual accuracy: Bhashini translation plus human review for certification-critical answers.
- Legal-risk false positives: confidence scores and official BIS links enable independent verification.



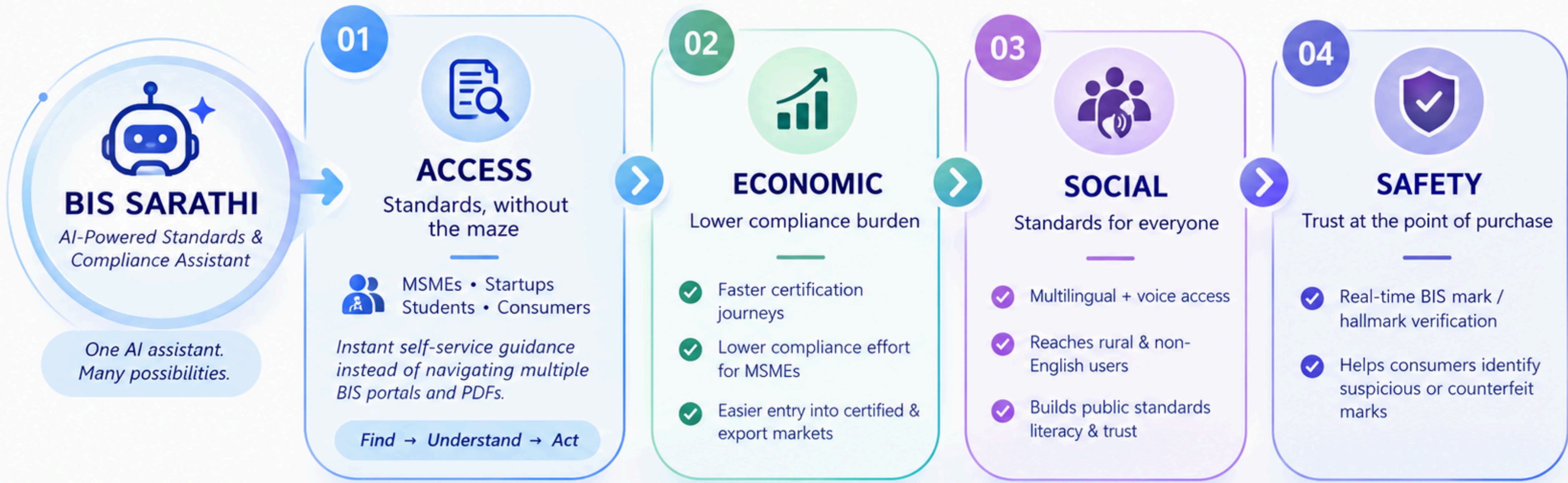
WHY IT SCALES

- Confidence-aware fallback: low-confidence queries escalate to human BIS helpdesk instead of guessing
- Reusable RAG core: new BIS schemes, standards and languages can be added as data modules—without rebuilding the core AI system.
- Open-source stack minimizes recurring licensing costs as usage grows.

From Data to Compliance

AI-powered assistant for BIS standards and regulatory compliance



**BIS SARATHI****ACCESS****ECONOMIC****INCLUSION****SAFETY**

Turning BIS's existing information infrastructure into accessible, actionable impact.

BIS ECOSYSTEM

BIS Official Portal — Indian Standards, certification & official services
<https://www.bis.gov.in>

Know Your Standard — Search IS number/keywords; standards, amendments, gazette notices, testing & labs

Compulsory Certification / QCOs — Mandatory compliance per Quality Control Orders

RESEARCH INSIGHTS

Compliance is contextual: Varies by product/QCO → Need: context-aware guidance

<https://www.bis.gov.in/product-certification/products-under-compulsory-certification/>

RAG ≠ guaranteed accuracy: Research shows hallucinations → Need: evidence & verification. (<https://arxiv.org/abs/2401.00396>)

Tools ≠ conversational: Discovery exists → Opportunity: intelligence layer
<https://arxiv.org/abs/2309.15217>

TECHNOLOGY RESEARCH

Retrieval-Augmented Generation (RAG) — Retrieves context for LLM Q&A
<https://arxiv.org/abs/2005.11401>

RAG Evaluation — Evaluates retrieval relevance, faithfulness & generation quality
<https://arxiv.org/abs/2405.07437>

Hallucination Risk — RAG produces unsupported claims → evidence + verification (<https://aclanthology.org/2025.cl-4.9/>)

Document AI & Multilingual — Structured OCR extraction & multi-language interaction (<https://arxiv.org/abs/2510.13366>)

KEY REFERENCES

BIS — Know Your Standard: Verification, standards & certification services.

BIS CARE App: Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks.

BIS — Enforcement Activities: Official BIS information on ISI mark misuse and enforcement. (<https://www.bis.gov.in/enforcement-activities>)

RAGAS: Automated Evaluation of Retrieval Augmented Generation.

RAGTruth: A Hallucination Corpus for Developing Trustworthy RAG Models.